

An Exact Collapse Threshold for Conserved-Resource Models of Protein Quality Control

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Supplementary Material

This document accompanies the paper above. Every number in the caption is recomputed by the script named in it and asserted by the accompanying test suite.

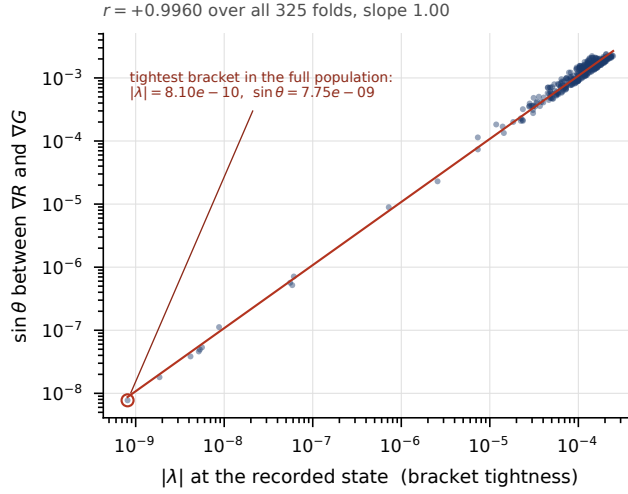


Figure S1: The parallelism residual is bracket tolerance, not failure of the identity. Over all 325 folds of the load grid, $\sin \theta$ between ∇R and ∇G tracks the leading eigenvalue recorded at the same state across five decades, with a log-log correlation of $+0.9960$ and a slope of 1.00 — the residual is *proportional* to how loosely the state is bracketed, which is what bracket tolerance means and what a failing identity would not produce. The circled point is the tightest bracket in the population, $|\lambda| = 8.10 \times 10^{-10}$ with $\sin \theta = 7.75 \times 10^{-9}$. Normalisation, reported here rather than given a panel of its own: on these same 325 states the max-normalised residual correlates with $|\lambda|$ at -0.835 and reaches 1.54×10^{-2} , while the gradient-normalised residual used throughout correlates at $+0.041$ and reaches 1.29×10^{-9} . The first degrades as the bracket tightens on the object it is meant to verify, because both $\det J$ and the cross product vanish at a saddle-node and their ratio is $0/0$ there; the second does not. Every number in this caption is recomputed by `scripts/figures/fig_identity.py:captionNumbers` and asserted by the test suite.